

Features	Open source database	Commercial database
Flexibility	Open source offers more flexibility to users	Proprietary software is less flexible and often comes with restrictions
Maintenance	Developed and maintained by a community	Developed and maintained by the organisation that published it
Support	May not have the support or capacity of a proprietary database	Single contact point for any problems that occur
Licences	Free to view, download, modify, distribute, and reuse	Does not allow access to the source code or modifications
Compatibility	No guarantee that opensource software will work in each user's environment	Compatible with the user's environment
Cost	Zero, upfront licensing costs	Expensive to license and maintain proprietary software
Access to data	With an open format, data sets can be returned from simple SQL query statements	Data cannot be accessed and viewed directly. Custom or proprietary tools must be used to access your data
Top databases	MySQL, PostgreSQL, MongoDB, Cassandra	Oracle, Microsoft SQL Server, DB2, Microsoft Access

Table 1: A comparison of open source and commercial databases

Open source databases are fully developed, just like databases. All the facilities and features that most people require are already a part of the product. Also, users can customise their open source database, and that is why it is very flexible to use.

And, of course, there is a huge community of users for the most popular open source products; so a wide range of documentation is available on them. In addition, there are forums where people can ask questions about these products to the experts.

Let us now take a brief look at the popular open source databases.

MySQL

MySQL was developed by Oracle Corporation and is written in C, C++ languages. Its stable version is 8.0.27 and the release date for that is October 19, 2021.

MySQL is one of the most popular and widely used SQL databases. It is also one of the top databases in Web applications. Some of the world's largest organisations (e.g., Facebook, Uber, Google, Adobe) use MySQL.

PostgreSQL

PostgreSQL is a powerful, and perhaps the world's most advanced open source database with over 30 years of active development. This database is well known for its reliability, feature robustness, and performance. PostgreSQL runs on all major operating systems including Windows, Mac, Linux and UNIX.

PostgreSQL databases provide enterprise-class database solutions and are used by a wide variety of enterprises across many industries

such as financial services, government, information technology, and media and communications.

MariaDB

MariaDB is a powerful database solution that is entirely free and supported by the MariaDB Foundation. It is powerful, versatile database software at no cost. MariaDB reads and writes 24 per cent faster as compared to MySQL. It also supports more connections than MySQL, while replication is 2x faster.

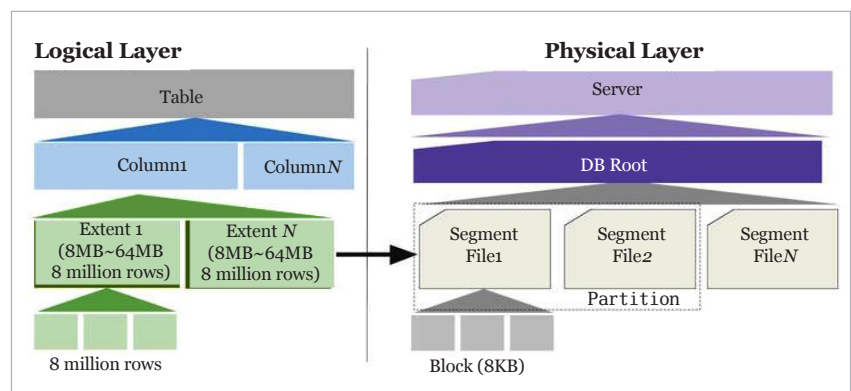


Figure 2: ColumnStore storage architecture (Reference: <https://mariadb.com/kb/en/columnstore-storage-architecture/>)